EXPERIMENT-2

Aim: Convert the ER diagram into a relational schema. Write complete MYSQL.CREATE TABLE statements with Primary Keys, foreign Keys, NOT NULL, UNIQUE, ON DELETE CASCADE and SETNULL. Constraints. Insert sample data and demonstrate referential integrity violations.

Source Code :-

1. Relational Schema

Table	Primary Key	Foreign Key (Imp)
CUSTOMER	(CustomerID)	—
CUSTOMER_PHONE	(CustomerID, Phone)	CustomerID
SELLER	(SellerID)	CustomerID
SELLER_PHONE	(SellerID, Phone)	SellerID
ADDRESS	(AddressID)	CustomerID
CATEGORY	(CategoryID)	ParentCategoryID
PRODUCT	(ProductID)	ProductID
ELECTRONICS	"	"
CLOTHING	"	"
BOOKS	"	"
ORDERS	(OrderID)	CustomerID, AddressID
ORDERITEM	(OrderID, ItemNo)	OrderID, ProductID
PAYMENT	(PaymentID)	OrderID
DELIVERY	(DeliveryID)	OrderID, AddressID

2. SQL Query

```
CREATE DATABASE ECOMMERCE;
USE ECOMMERCE;

CREATE TABLE CUSTOMER(
    CustomerID INT PRIMARY KEY,
    Name VARCHAR(50) NOT NULL,
    Email VARCHAR(100) NOT NULL,
    Password VARCHAR(100) NOT NULL,
    DateOfBirth DATE NOT NULL);

CREATE TABLE CUSTOMER_PHONE(
    CustomerID INT,
    Phone VARCHAR(15),
    PRIMARY KEY(CustomerID, Phone)
    FOREIGN KEY(CustomerID) REFERENCES CUSTOMER
    (CustomerID) ON DELETE CASCADE);

CREATE TABLE SELLER(
    SellerID INT PRIMARY KEY,
    SellerName VARCHAR(50) NOT NULL,
    Email VARCHAR(100) UNIQUE NOT NULL,
    GSTIN VARCHAR(20) UNIQUE NOT NULL,
    AccountNo VARCHAR(20),
    IFSCCode VARCHAR(15),
    BankName VARCHAR(50));
```

```
CREATE TABLE SELLER_PHONE (
    SellerID INT,
    Phone VARCHAR(15),
    PRIMARY KEY (SellerID, Phone),
    FOREIGN KEY (SellerID) REFERENCES SELLER
    (SellerID) ON DELETE CASCADE);

CREATE TABLE CATEGORY (
    CategoryID INT PRIMARY KEY,
    CategoryName VARCHAR(50) NOT NULL,
    Description VARCHAR(100),
    ParentCategoryID INT,
    FOREIGN KEY (ParentCategoryID) REFERENCES CATEGORY
    (CategoryID) ON DELETE SET NULL);

CREATE TABLE PRODUCT (
    ProductID INT PRIMARY KEY,
    ProductName VARCHAR(50) NOT NULL,
    Description VARCHAR(100),
    Price DECIMAL(10, 2) NOT NULL,
    StockQty INT NOT NULL,
    Weight DECIMAL(8, 2),
    SellerID INT NOT NULL,
    CategoryID INT,
    FOREIGN KEY (SellerID) REFERENCES SELLER
    (SellerID) ON DELETE CASCADE,
    FOREIGN KEY (CategoryID) REFERENCES CATEGORY
    (CategoryID) ON DELETE SET NULL);
```

```
CREATE TABLE ELECTRONICS (
    ProductID INT PRIMARY KEY,
    Brand VARCHAR(50),
    Warranty VARCHAR(30),
    PowerRating VARCHAR(30),
    FOREIGN KEY (ProductID) REFERENCES PRODUCT(ProductID)
    ON DELETE CASCADE);

CREATE TABLE CLOTHING (
    ProductID INT PRIMARY KEY,
    Size VARCHAR(20),
    Colour VARCHAR(30),
    Material VARCHAR(30),
    FOREIGN KEY (ProductID) REFERENCES PRODUCT(ProductID)
    ON DELETE CASCADE);

CREATE TABLE BOOKS (
    ProductID INT PRIMARY KEY,
    ISBN VARCHAR(20) UNIQUE,
    Author VARCHAR(50),
    Publisher VARCHAR(50),
    FOREIGN KEY (ProductID) REFERENCES PRODUCT(ProductID)
    ONDELETE CASCADE);
```

```
CREATE TABLE ADDRESS(
    AddressID INT PRIMARY KEY,
    DoorNo VARCHAR(10),
    Street VARCHAR(50),
    Area VARCHAR(50),
    City VARCHAR(30),
    State VARCHAR(30),
    Pincode VARCHAR(10) NOT NULL,
    AddressType VARCHAR(20),
    CustomerID INT NOT NULL
    FOREIGN KEY (CustomerID) REFERENCES CUSTOMER
    (CustomerID) ON DELETE CASCADE);

CREATE TABLE ORDERITEM(
    OrderID INT,
    ItemNo INT,
    ProductID INT NOT NULL,
    ItemPrice DECIMAL(10,2) NOT NULL,
    Quantity INT NOT NULL,
    Discount DECIMAL(10,2),
    TaxAmount DECIMAL(10,2),
    PRIMARY KEY(OrderID, ItemNo),
    FOREIGN KEY(OrderID) REFERENCES ORDERS
    (OrderID) ON DELETE CASCADE,
    FOREIGN KEY (ProductID) REFERENCES RRODUCT
    (ProductID) ON DELETE CASCADE);
```

```
CREATE TABLE PAYMENT (
    PaymentID INT PRIMARY KEY,
    PaymentDate DATE NOT NULL
    Amount DECIMAL(10,2) NOT NULL,
    PaymentMethod VARCHAR(30) UNIQUE,
    TransactionID VARCHAR(30) UNIQUE,
    PaymentStatus VARCHAR(20) NOT NULL,
    OrderID INT UNIQUE NOT NULL
    FOREIGN KEY(OrderID) REFERENCES ORDERS
    (OrderID) ON DELETE CASCADE);

CREATE TABLE DELIVERY (
    DeliveryID INT PRIMARY KEY,
    DeliveryDate DATE,
    DeliveryStatus VARCHAR(20) NOT NULL,
    TrackingID VARCHAR(30) UNIQUE,
    DeliveryPartner VARCHAR(50),
    ShippingCharge DECIMAL(8,2),
    OrderID INT UNIQUE NOT NULL,
    AddressID INT,
    FOREIGN KEY(OrderID) REFFRENCES ORDERS
    (OrderID) ON DELETE CASCADE,
    FOREIGN KEY(AddressID) REFERENCES ADDRESS
    (AddresID) ON DELETE SET NULL);
```

```
INSERT INTO CUSTOMER VALUES (
1, 'Ankan', 'ankan@gmail.com', 'pass123', 2006-07-20);

INSERT INTO CUSTOMER_PHONE VALUES(
1, '9215613412');

INSERT INTO SELLER VALUES
(101, 'Techstore', 'tech@gmail.com', 'GST123', '123456', 'SBIN001', 'SBI')

INSERT INTO SELLER_PHONE VALUES
(101, '8131341161');

INSERT INTO CATEGORY VALUES
(201, 'Electronics', 'Electronic items', NULL);

INSERT INTO PRODUCT VALUES
(301, 'Laptop', 'Graming Laptop', 75000, 10, 2.5, 101, 201);

INSERT INTO ADDRESS VALUES
(401, '12', 'Barrackpore', 'North 24 PGs', 'Kolkata', 'West Bengal',
'700122', 'Home', 1);

INSERT INTO ORDERS VALUES
(501, '2026-08-14', 'Placed', 75000, 1, 401);

INSERT INTO ORDERITEM VALUES
(501, 1, 301, 75000, 1, 1000, 9000);

INSERT INTO PAYMENT VALUES
(601, '2026-08-14', 74000, 'UPI', 'TXN123', 'Paid', 501);

INSERT INTO DELHIVERY VALUES
(701, NULL, 'Processing', 'TRK123', 'Delhivery', 100, 501, 401);
```

Output :-

1) Show Tables

```
SHOW TABLES;
```

Tables_in_ecommerce
ADDRESS
BOOKS
CATEGORY
CLOTHING
CUSTOMER
CUSTOMER_PHONE
DELIVERY
ELECTRONICS
ORDERS
ORDERITEM
PAYMENT
PRODUCT
SELLER
SELLER_PHONE

2) Sample Data

```
SELECT * FROM CUSTOMER;
SELECT * FROM ORDERS;
```

Customer

CustomerID	Name	Email	Password	DateOfBirth
1	Ankan	ankan@gmail.com	Pass123	2006-07-20

OrderID	ORDERDATE	OrderStatus	TotalAmount	CustomerID	AddressID
501	2026-08-14	Placed	75000.00	1	401

3) Referential integrity violation

```
INSERT INTO ORDER VALUES
(502, '2026-08-14', 'Closed', 50000, 999, 401);
```

-> ERROR 1452 (23000):
Cannot add or update a child row:
a foreign key constraint fails

Reason - CustomerID 999 does not exist in Customer Table

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